POTW Deployment Guide

A click-by-click guide for Namecheap, Resend, Render, and PostgreSQL

Prepared for the Purdue Math Club Problem of the Week app

Security rule: Never paste a database password, Resend API key, JWT secret, or complete connection URL into GitHub, screenshots, Discord, or chat. Put secrets only in Render environment variables.

Contents

1 What you are deploying

The repository is a single project (a monorepo), but Render will run two services from it:

- **Static Site:** builds the React files in `src/` and serves the website.
- **Web Service:** runs the Express API in `server/`. It handles login, email, submissions, grading, hints, and database access.
- **PostgreSQL:** stores users, problems, submissions, grades, seasons, and hint requests.

The request path is: browser → React Static Site → Express Web Service → PostgreSQL.

2 Before clicking anything

Have these ready:

1. A GitHub repository containing `src/`, `server/`, `db/`, `public/`, and `package.json`.
2. A Namecheap account and the domain `purdue-math-potw.com`.
3. A Resend account.
4. A Render account connected to the GitHub account.
5. PostgreSQL command-line tools (`psql`) installed locally for one-time migrations.

The repository must not contain `.env`, `node_modules/`, or `dist/`.

3 Save and push the project

Open PowerShell and run:

```
cd "C:\textbackslash Users\textbackslash YOUR\_NAME\textbackslash Documents\textbackslash GitHub\textback
git add .
git commit -m "Prepare POTW for deployment"
git push origin main
```

If Git asks for a login, complete the GitHub sign-in. If the push fails, stop and fix that before continuing.

4 Namecheap and Resend email setup

Step Create the Resend domain entry

1. Go to https://resend.com and sign in.
2. In the left menu click Domains.
3. Click Add Domain.
4. Enter `purdue-math-potw.com` and click Add.
5. Leave this Resend page open. It displays the DNS records you must copy.

Step Add DNS records in Namecheap

1. Go to https://www.namecheap.com and sign in.
2. Click Domain List in the left sidebar.
3. Find purdue-math-potw.com and click Manage.
4. Click the Advanced DNS tab.
5. Under Host Records, click Add New Record.
6. For every record shown by Resend, choose the same type and paste its exact Host and Value.
7. Set TTL to Automatic, then click the green checkmark/save icon.

For a Namecheap Host field, use @ when Resend means the root domain. If Resend gives a full hostname, Namecheap may only need the part before .purdue-math-potw.com. Do not guess: use Resend's displayed values. Do not create a second SPF record if one already exists.

Step Verify in Resend

1. Return to Resend Domains.
2. Click purdue-math-potw.com.
3. Click Verify DNS Records.
4. Wait until the domain status says Verified.

Step Create the Resend API key

1. In Resend click API Keys.
2. Click Create API Key.
3. Name it POTW Production.
4. Select Sending access and restrict it to purdue-math-potw.com when offered.
5. Click Create Key.
6. Copy the complete key beginning with re_. Resend only displays it once.

5 Create Render PostgreSQL

1. Go to https://dashboard.render.com.
2. Click New in the upper-right corner.
3. Select PostgreSQL.
4. Name it potw-db.
5. Choose the same region you will use for the backend (Ohio is appropriate for Purdue users).
6. Choose a paid production plan if you need reliable always-on production behavior.

7. Click `Create Database`.
8. Wait until the database status says `Available`.
9. Open the database, click `Connect`, and copy the **External Database URL** for local migrations.

Use the database's **Internal Database URL** later in the Render backend service. Render recommends internal connections for services in the same region.

6 Unverified signup behavior

The app intentionally does not let an abandoned, unverified signup reserve a username or email forever. When a new registration matches an existing **unverified** username or email, the old unverified account is replaced and a fresh verification email is sent. A verified account is never replaced; it correctly receives a "username or email already registered" error. This also invalidates the old verification link, so use only the newest email link.

To test this after deployment:

1. Register a username and email, but do not click its verification link.
2. Register again with the same username and email.
3. Confirm the second registration succeeds and a new email arrives.
4. Click only the newest link; the old link should be invalid.
5. Verify the account and then try registering the same username again; this time it must be rejected.

7 Run the database migrations

These commands create the schema and apply all features. Run them once, in this exact order.

Step Open PowerShell in the repository

```
cd "C:\textbackslash Users\textbackslash YOUR\_NAME\textbackslash Documents\textbackslash GitHub\textback
$prodDbUrl = Read-Host "Paste the Render External Database URL"
```

Paste the URL only when PowerShell prompts you. It will not be printed in this guide.

Step Run each file

```
psql $prodDbUrl -f db/001_schema.sql
psql $prodDbUrl -f db/002_sample_problems.sql
psql $prodDbUrl -f db/003_admin_features.sql
psql $prodDbUrl -f db/004_submission_options.sql
psql $prodDbUrl -f db/005_account_security.sql
psql $prodDbUrl -f db/006_problem_release_times.sql
psql $prodDbUrl -f db/007_problem_due_times.sql
psql $prodDbUrl -f db/008_default_due_time.sql
psql $prodDbUrl -f db/009_align_seed_season.sql
psql $prodDbUrl -f db/010_hint_responses.sql
```

Step Verify

```
psql $prodDbUrl -c "\\dt"
psql $prodDbUrl -c "SELECT COUNT(*) AS problems FROM problems; SELECT COUNT(*) AS seasons FROM seasons;"
```

You should see tables including `users`, `problems`, `submissions`, `seasons`, `problem_proposals`, and `hint_requests`.

8 Create the Render backend

1. In Render click New → Web Service.
2. Select the GitHub repository and click Connect.
3. Leave Root Directory blank. The project is now at the repository root.
4. Set Runtime to Node.
5. Set Build Command to `npm install`.
6. Set Start Command to `npm run start:server`.
7. Click Create Web Service.

Step Add backend environment variables

Open the new service, click Environment in the sidebar, then click Add Environment Variable. Add each row:

`NODE_ENV`	`production`
`DATABASE_URL`	Render PostgreSQL **Internal Database URL**
`JWT_SECRET`	A new random value generated locally
`FRONTEND_URL`	Temporarily `https://example.com`
`TRUST_PROXY`	`1`
`SMTP_HOST`	`smtp.resend.com`
`SMTP_PORT`	`587`
`SMTP_SECURE`	`false`
`SMTP_USER`	`resend`
`SMTP_PASS`	Complete Resend API key
`EMAIL_FROM`	`Purdue Math Club POTW <noreply@purdue-math-potw.com>`

Generate the JWT value in PowerShell:

```
[Convert]::ToBase64String([Security.Cryptography.RandomNumberGenerator]::GetBytes(48))
```

Click Save Changes; Render redeploys the service.

Step Test the backend

Copy the backend's Render URL, then open:

```
https://YOUR-BACKEND.onrender.com/api/health
```

The page must display `{"status":"ok"}` before you create the frontend.

9 Create the Render frontend

1. In Render click New → Static Site.
2. Select the same GitHub repository and click Connect.
3. Leave Root Directory blank.
4. Set Build Command to `npm install && npm run build`.
5. Set Publish Directory to `dist`.
6. Under Environment Variables add `VITE_API_URL`. Set its value to the backend URL:

```
VITE_API_URL=https://YOUR-BACKEND.onrender.com
```

7. Click Create Static Site.

Step Add the SPA rewrite

In the frontend service click Redirects/Rewrites (or Settings → Redirects and Rewrites), click Add Rule, and enter:

Source	Destination	Action
`/*`	`/index.html`	`Rewrite`

Save the rule and redeploy if Render asks.

10 Connect the two services

Copy the frontend URL, for example `https://potw-frontend.onrender.com`. Return to the backend service, open Environment, and replace the temporary value with:

```
FRONTEND_URL=https://YOUR-FRONTEND.onrender.com
```

Save and redeploy the backend. This allows browser requests from the frontend to reach the API.

11 First production test

Run these tests in order:

1. Open the frontend Render URL.
2. Register a test account.
3. Confirm the verification email arrives.
4. Log in and submit a solution.
5. Log in as the admin and open Admin → Grading.
6. Grade the submission and confirm the leaderboard updates.
7. Open Hints, request a hint, and respond from the admin proposals tab.

8. Confirm the solver sees the response.
9. Test file upload, profile history, password reset, and anonymous mode.
10. Refresh `/leaderboard`, `/profile`, and `/hints`; each must still load because of the rewrite.

12 Common problems

No Environment tab	You opened the database or Static Site. Open the backend **Web Service**.
Emails fail with 403	The domain is not verified, the sender does not use the verified domain, or you used a shortened API key.
CORS error	Set backend `FRONTEND_URL` to the exact frontend URL, save, and redeploy.
Frontend API errors	Check `VITE_API_URL`; it must be the backend URL with no trailing `/api`.
Refresh gives 404	Add the Static Site rewrite `/*` → `/index.html` as a Rewrite.
Database connection fails	Use the Render Internal URL in the backend and confirm the database is Available.
Migrations fail with relation exists	Usually safe; many migrations are idempotent. Read the first actual ERROR line.

13 After launch

- Enable paid PostgreSQL backups/recovery appropriate for your data.
- Rotate the database credential and Resend key if either was exposed.
- Add a custom website domain later; Render's generated URL works without one.
- Keep the backend service running continuously because its scheduler archives expired problems.
- Never put secrets in frontend variables; anything beginning with `VITE_` is visible in the browser build.

14 Useful official documentation

- Render deployment: https://render.com/docs/your-first-deploy
- Render Postgres: https://render.com/docs/postgresql-creating-connecting
- Resend domains: https://resend.com/docs/dashboard/domains/introduction
- Resend API keys: https://resend.com/docs/dashboard/api-keys/introduction
- Resend SMTP: https://resend.com/docs/send-with-smtp